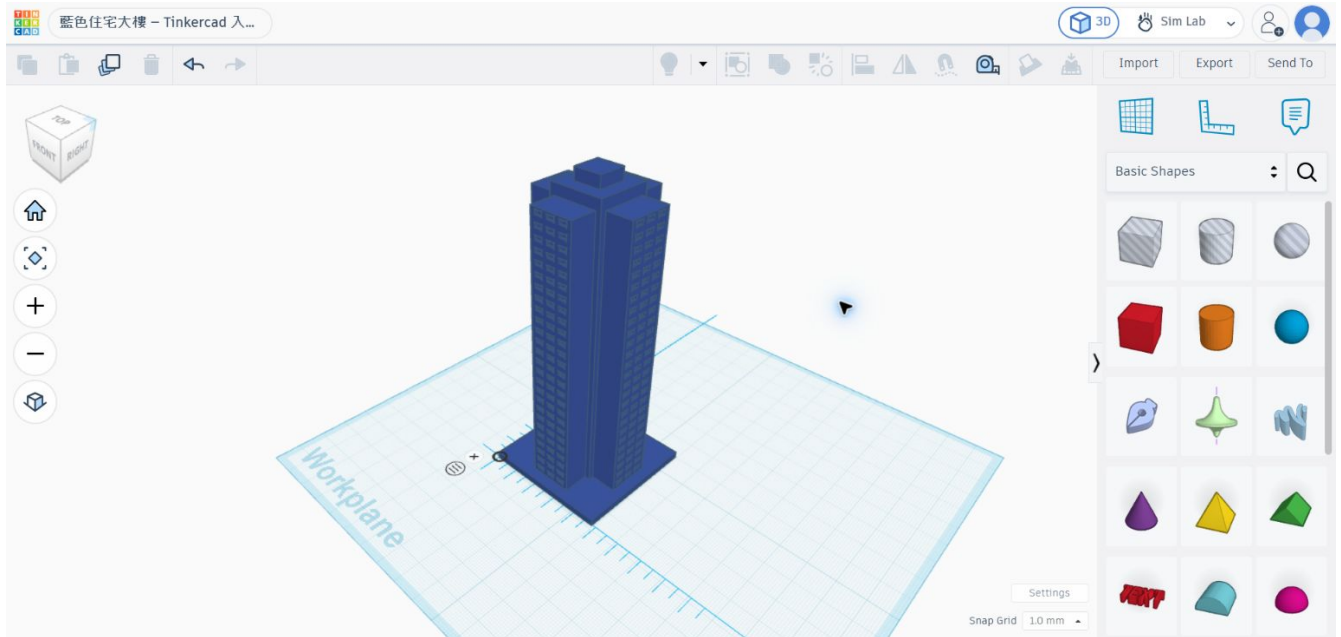


Blue Residential Tower | Student Guide

Form 2 STEAM • Tinkercad 3D modelling



01 | Practise the basics

Mouse, keyboard, selection, dimensions and tool locations.

02 | Follow the screenshots

34 modelling steps, each with an action and a check.

03 | Export for your teacher

Check connections and size, download STL, and prepare for 3D printing.

How to use this guide

Practise the basic functions first. Finish each page before moving on.

Stage 1 | Basic controls

Use the practice design to try the mouse, keyboard, dimensions and grouping. Find the button, do the action, then check the result.

Stage 2 | Basic building

Steps 01-18 make the base, tower, four wings, plant room and blue finish. All dimensions use millimetres (mm).

Stage 3 | Windows and export

Steps 19-30 add windows; steps 31-34 check, export STL and submit. If time is short, submit the basic version first.

Remember three things

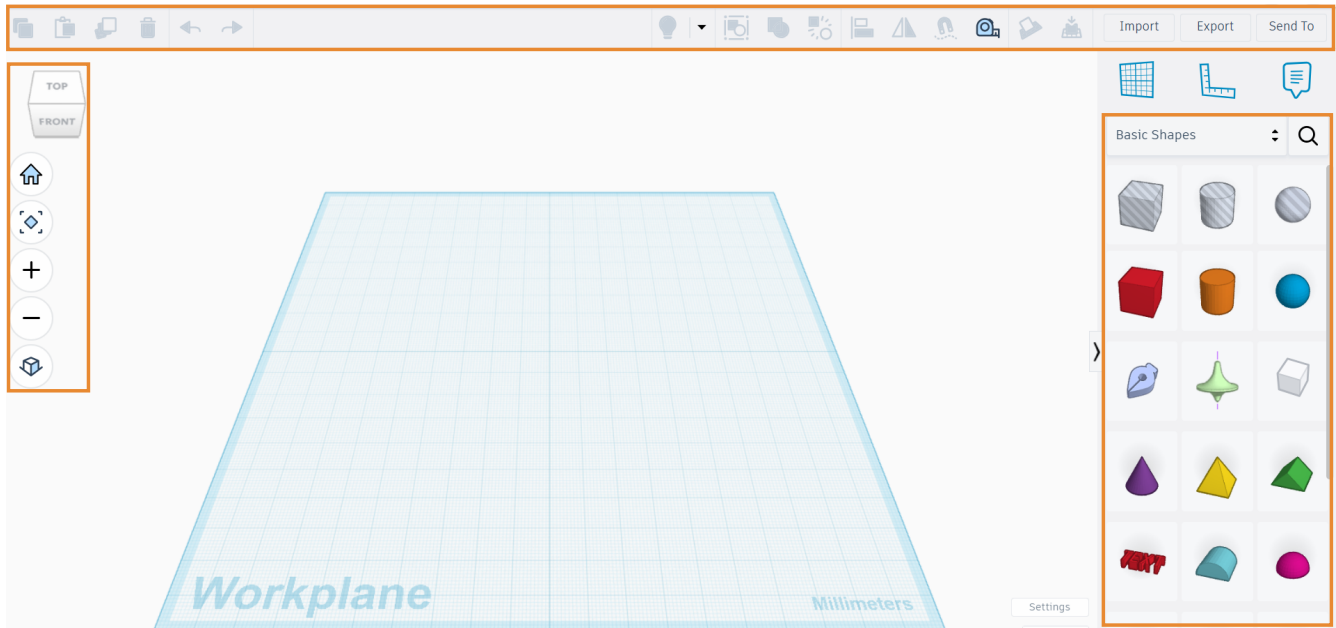
Press Enter after every number. Press Ctrl+Z after a mistake. Press Fit all in view if the model disappears.

Reading screenshots

Screenshots come from an earlier Tinkercad build. The interface may change slightly; hover over an icon to read its English name. This guide does not teach position coordinates.

Meet the Tinkercad editor

Orange outlines mark the tool areas and keep the text readable.



Left | View controls

ViewCube switches TOP and FRONT; Home returns to the default angle; Fit all in view puts the model on screen.

Top | Main toolbar

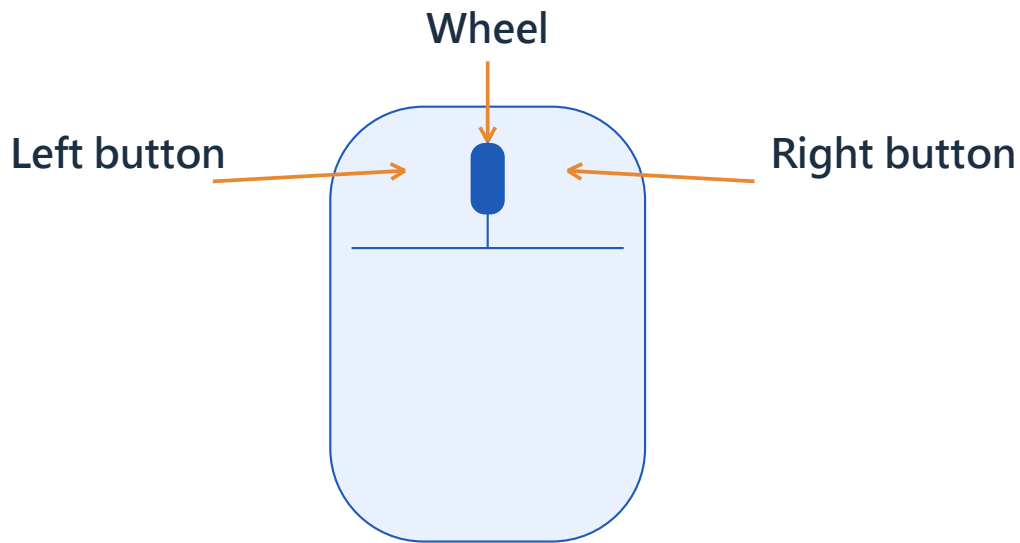
Undo, Duplicate, Align and Group become available when the correct objects are selected.

Centre and right | Workplane and shapes

The blue grid is the Workplane. Basic Shapes contains solid red Box and striped Hole.

Mouse: identify the three controls

The arrow tip points to the button to press. The drawing uses a standard left and right mouse button.



Left button: select and drag

Click an object once to select it. Hold the left button, move the mouse, then release to drag. Click empty space to deselect.

Wheel: zoom the view

Scroll forward to zoom in and backward to zoom out. This changes the view only; the model size stays the same.

Right button: orbit the view

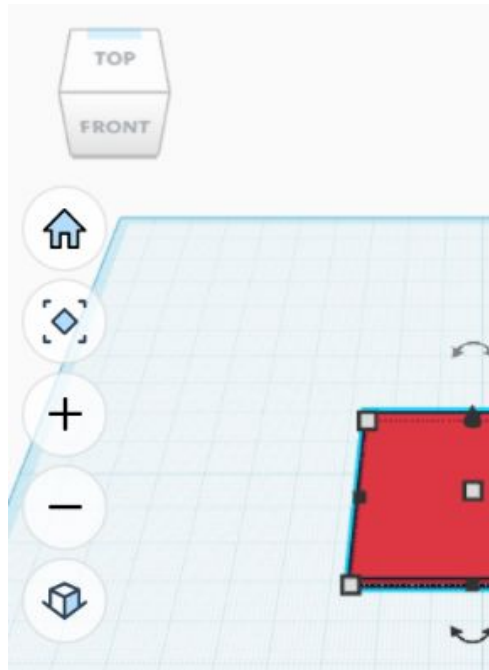
Place the pointer in the work area, hold the right button and drag to see another side of the building.

Shift + right button: pan

Hold Shift, then hold the right button and drag. Release both when the view is in place.

View the model: orbit and recover it

Practise view controls; do not drag the model itself.



1 | Click TOP, then FRONT

TOP shows the plan view; FRONT shows the front elevation. Click Home to return to the default angle.

2 | Fit all, then + / -

The four-corner icon is Fit all in view. The + and - controls change viewing distance.

3 | Try right button and Shift + right button

Right-drag orbits the view. Shift + right-drag pans the view. The model dimensions do not change.

Keyboard 1: control keys and delete

Orange outlines mark the key edges; enlarged views are shown below.



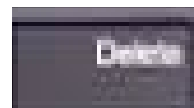
Ctrl



Shift



Enter



Delete



Backspace

How to use Ctrl combinations

Hold Ctrl, tap the second key, then release both. For example, Ctrl+Z. Click the work area before testing a shortcut.

Shift, Enter and Delete

Shift adds objects to a selection; Enter confirms a number; Delete or Backspace removes the selected object.

Mini practice | delete and recover

Select a practice Box, press Delete, then press Ctrl+Z. Check that the Box returns.

Keyboard 2: common letter shortcuts

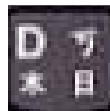
Orange outlines mark the key edges; enlarged views are shown below.



Z



A



D



G



H

Ctrl+Z undo; Ctrl+D duplicate

Undo a mistake. A duplicate starts on top of the original, so move it away before checking it.

Ctrl+A select all; Ctrl+G group

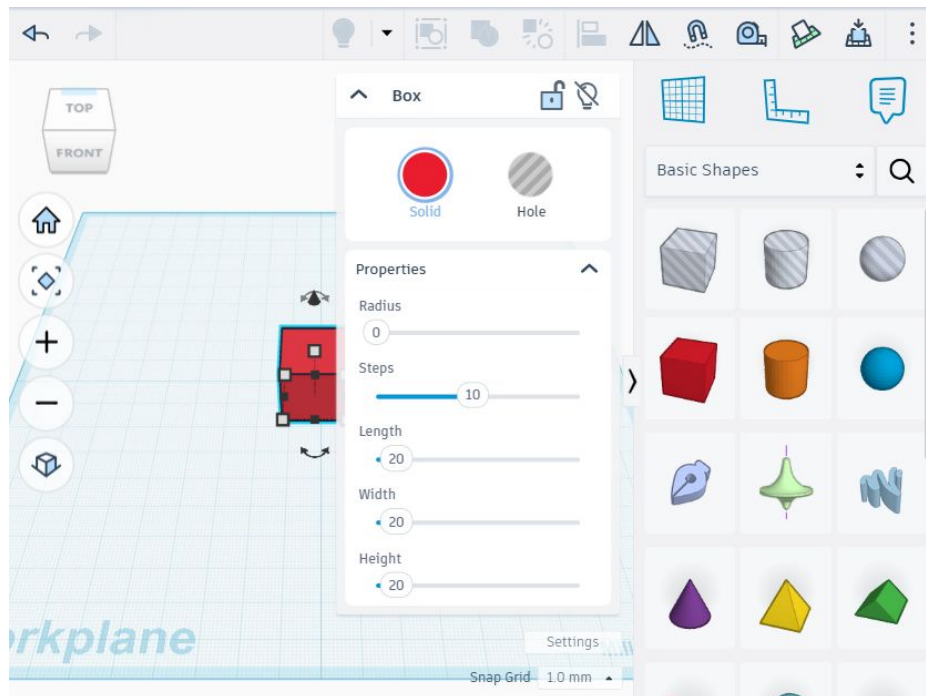
Check what is on screen before selecting all. Ctrl+G combines selected solids; a solid and a Hole together make a cut.

Ctrl+H hide; D drop to Workplane

Ctrl+H hides a selected object. Show all brings it back. D drops an object onto the current Workplane; Ctrl+D duplicates.

Try once: drag in and select a Box

Work in the practice design; begin the real tower after this page.



1 | Drag in a solid Box

Find the red Box in Basic Shapes, hold the left button, drag it to the blue grid and release.

2 | Select and deselect

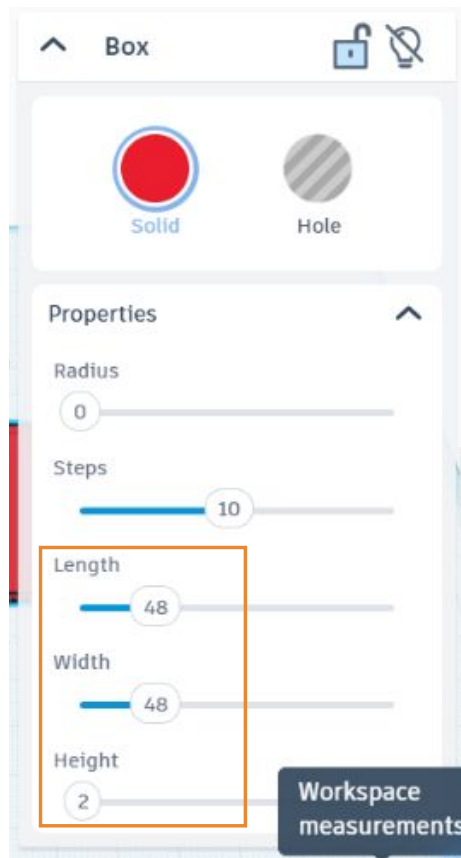
Click the Box once to see its outline and handles. Click empty space to remove the handles.

3 | Select two objects

Add another Box. Click the first, hold Shift and click the second. Both should show a selection outline.

Change size: length, width and height

The screenshot enlarges the Box panel so the numbers are easy to find.



1 | Select Box and open Properties

Select the Box and find Properties in the Shape panel. Click the small arrow if it is collapsed.

2 | Enter 48, 48 and 2

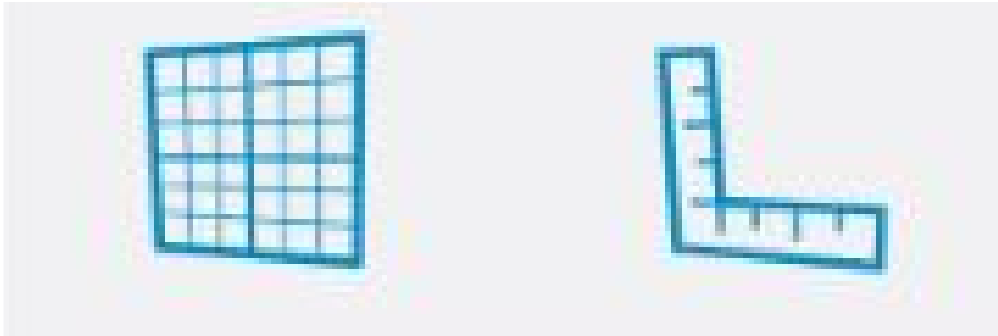
Length is front-to-back, Width is left-to-right and Height is the object height. Click a number, type it, then press Enter.

3 | Check

The Box should become a thin plate. Do not confuse the black lift handle with the height field.

Place on a surface: Workplane and D

Use a surface Workplane; do not enter position coordinates.



1 | Choose a support surface

Click the blue Workplane grid tool, then click the top of the base to place a temporary Workplane.

2 | Drop the object onto the surface

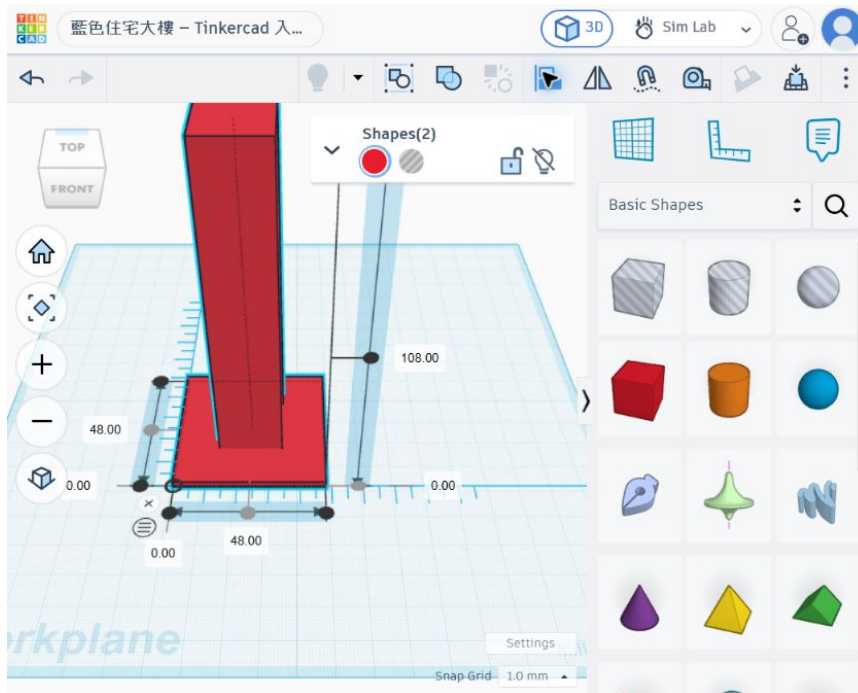
Drag in a new Box, select it and press D. The Box should touch the top surface of the base.

3 | Return to the ground grid

Click Workplane again and click an empty grid area beside the model. Use the same method on the tower top.

Align: select two objects first

This is the key step for centring the tower on the base.



1 | Shift-click the base and tower

Click empty space, hold Shift and click the two objects. Click Align or press L.

2 | Click the two horizontal centre dots

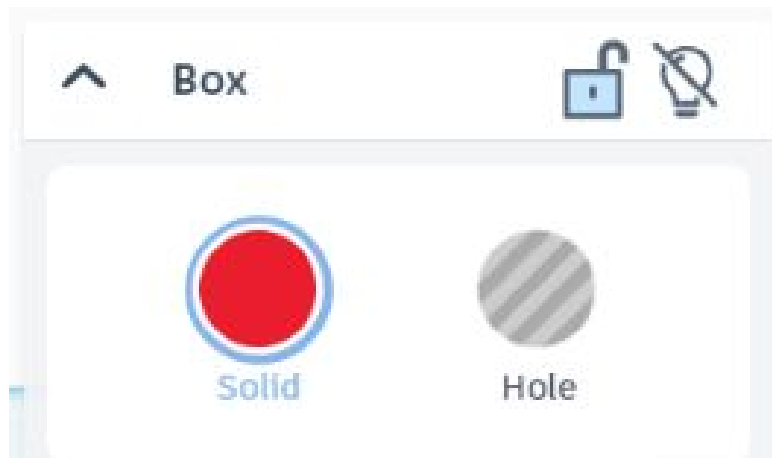
Click the centre dot for left-right and the centre dot for front-back. Do not click the vertical centre dot.

3 | Check

From TOP, the tower has equal space on opposite sides. From FRONT, its bottom still touches the base.

Holes and groups: making a window recess

Try it on a practice Box; do not change the finished tower yet.



1 | Solid and Hole

The red circle is Solid; the striped circle is Hole. Select a small Box and click Hole.

2 | Overlap before grouping

Move the Hole onto the solid so they overlap. Hold Shift to select both, then press Ctrl+G (Union).

3 | Check and undo

The overlapping part is removed. A Hole beside the wall does nothing. Ctrl+Z returns to before the group.

Before official modelling: preparation check

Finish the basic practice, then open a new design.

Name and units

Rename the design Class_Name_BlueResidentialTower. In Settings / Edit Grid, choose millimetres; a 200 × 200 mm grid is enough.

Target dimensions

Base 48 × 48 × 2 mm; tower 20 × 20 × 108 mm; plant room height 4.3 mm. Total height with base: 114.3 mm.

Scale

80 m = 80,000 mm. $80,000 \div 700$ is about 114.3 mm, so this example is about 1:700. It is a teaching model, not a survey drawing.

Working setup

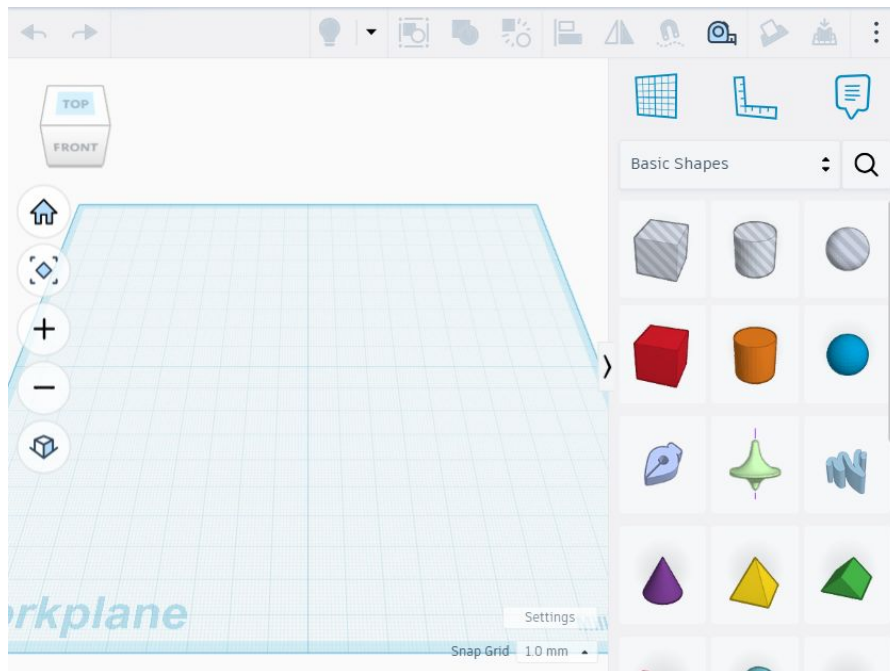
This guide shows a Windows keyboard and a standard mouse. Mac users can use the matching Command shortcut or the on-screen tool.

Try three actions

Try Ctrl+Z, Shift selection and Workplane + D once each. Then start model step 01.

01 Open a blank workplane

Model build step 1 / 34 | Workplane



Tinkercad screenshot - observe the tool and the model shape.

Find this

The Tinkercad start screen and the blue Workplane.

Do this

Follow the class link or the teacher's sign-in method. Choose Create > 3D Design to open a blank editor.

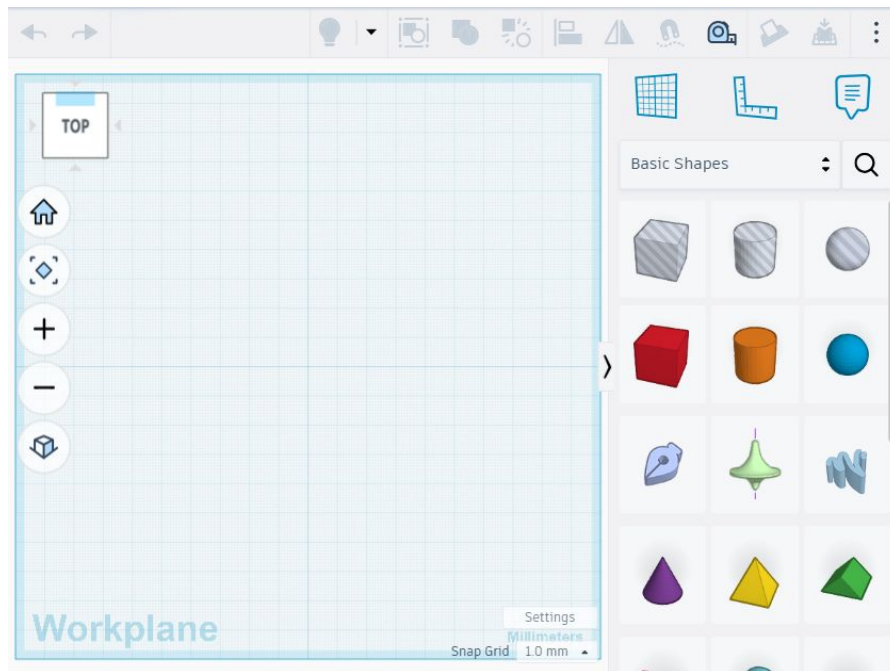
Check

The blue Workplane is in the centre and Basic Shapes is on the right.

Help: Keep the PDF beside Tinkercad. Ask the teacher about the class sign-in method.

02 Practise changing the view

Model build step 2 / 34 | ViewCube



Tinkercad screenshot - observe the tool and the model shape.

Find this

The ViewCube and the left viewing controls.

Do this

Click TOP and FRONT on the ViewCube, then click Home to return to the default angle.

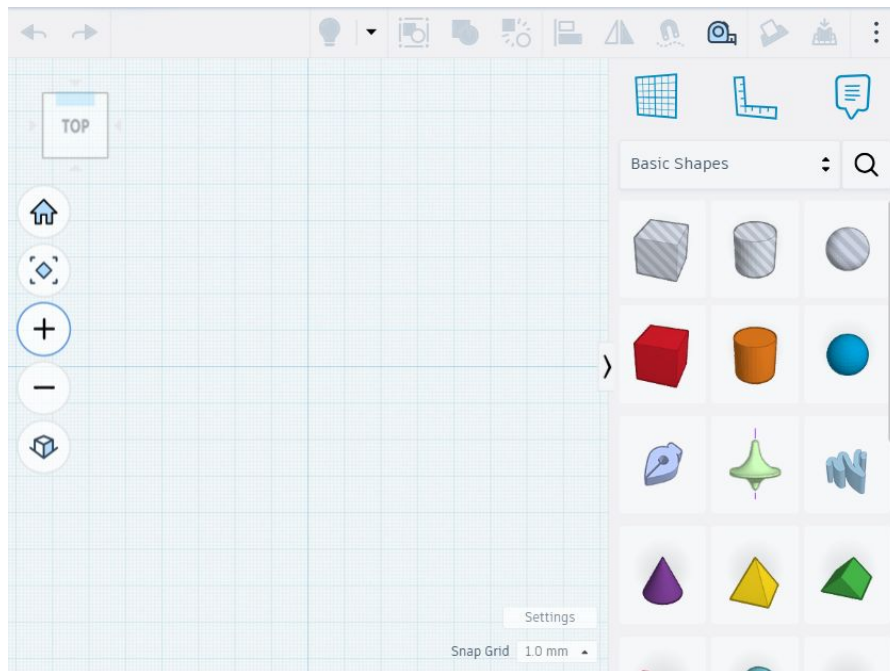
Check

You can return to the default view without moving the model.

Help: If the interface looks different, hover over an icon to read its English name.

03 Practise zooming

Model build step 3 / 34 | Zoom



Tinkercad screenshot - observe the tool and the model shape.

Find this

The zoom controls on the left side of the editor.

Do this

Use + and - or the mouse wheel to zoom. The model is not resized.

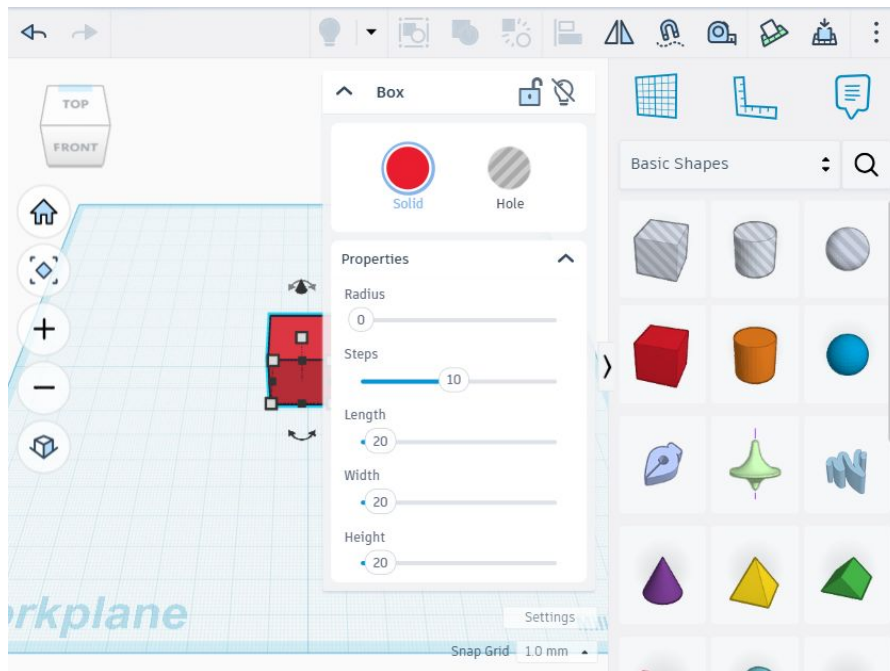
Check

The view changes size but the object dimensions do not.

Help: Fit all in view (the four-corner icon) brings a lost model back into view.

04 Drag in the first solid box

Model build step 4 / 34 | Box



Tinkercad screenshot - observe the tool and the model shape.

Find this

Basic Shapes on the right and the centre Workplane.

Do this

Drag a red Box from Basic Shapes onto the Workplane and release it.

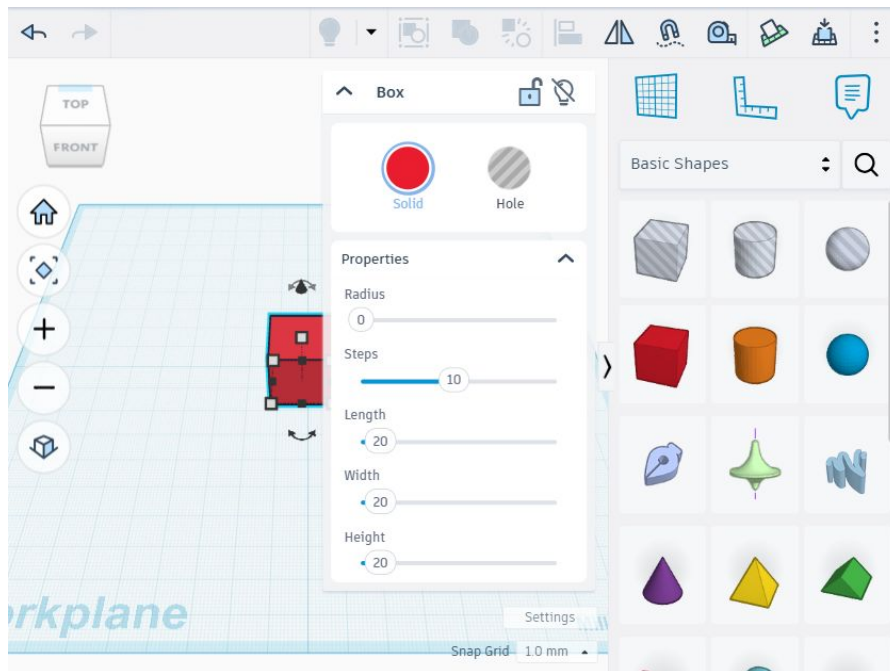
Check

A red Box sits on the Workplane.

Help: Do not drag the striped Hole by mistake; use the solid red Box for this step.

05 Learn selection and size handles

Model build step 5 / 34 | Workplane



Tinkercad screenshot - observe the tool and the model shape.

Find this

A selected Box and its white size handles.

Do this

Click the Box once. Notice the white handles and the height handle; do not drag them yet.

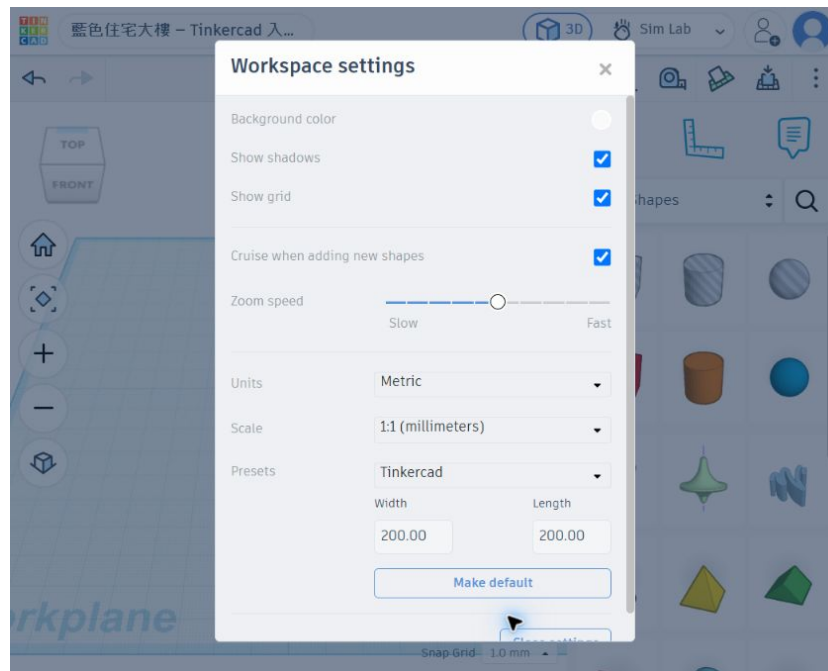
Check

The Box is selected and its handles are visible.

Help: White handles change size; the black lift handle changes height above the Workplane.

06 Use millimetres and understand scale

Model build step 6 / 34 | Grid settings



Tinkercad screenshot - observe the tool and the model shape.

Scale: about 1:700; Target total height: 114.3 mm; Calculation: $80,000 \div 700$

Find this

Grid settings and the Units menu.

Do this

Open Settings, choose Metric, keep the scale at 1:1 millimetres, and set Width and Length to 200. Close the settings panel.

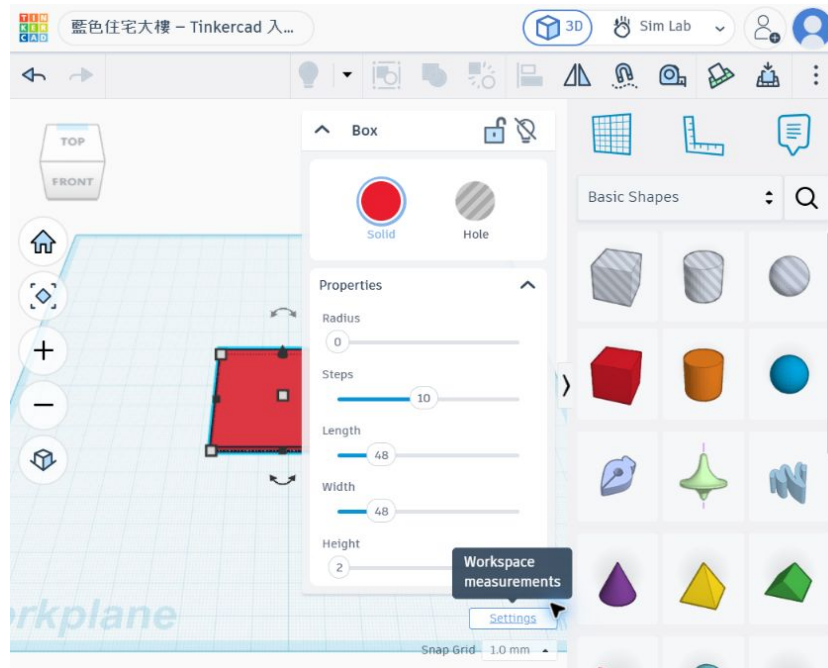
Check

All following entries use mm; the example total height is 114.3 mm, about a 1:700 model of an 80 m building.

Help: If a number is not accepted, click the field again and press Enter.

07 Turn the box into a thin base

Model build step 7 / 34 | Workplane



Tinkercad screenshot - observe the tool and the model shape.

Find this

The first Box on the Workplane.

Do this

Select the first Box and enter Width 48, Length 48 and Height 2 mm. Press Enter after every number.

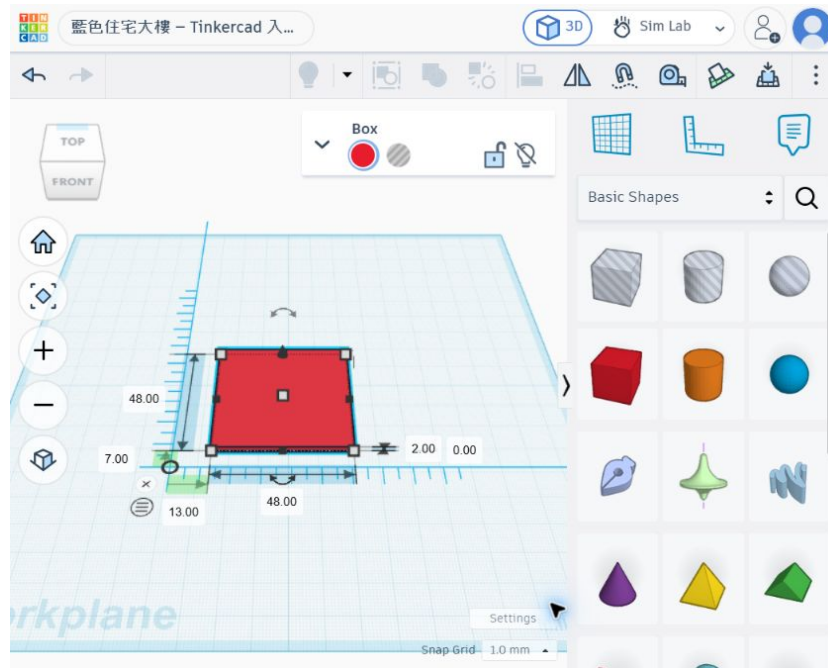
Check

A $48 \times 48 \times 2$ mm thin base is visible.

Help: A thin base is 2 mm high; do not make it thick by dragging a view control.

08 Read the object size with Ruler

Model build step 8 / 34 | Ruler



Tinkercad screenshot - observe the tool and the model shape.

Find this

The Ruler tool and the selected base.

Do this

Drag Ruler to an empty part of the Workplane, then select the base. Read its 48, 48 and 2 mm dimensions.

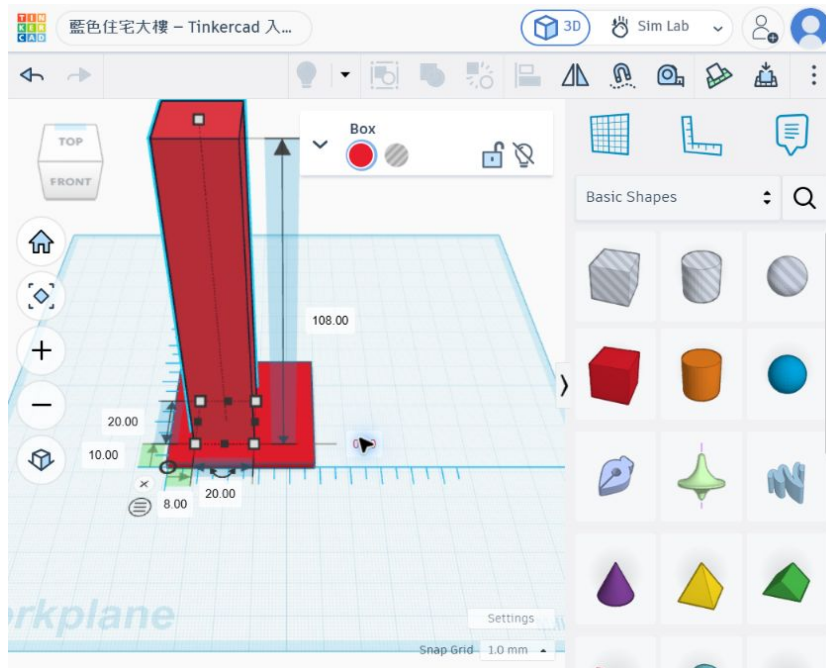
Check

Ruler shows the three base dimensions.

Help: Ruler reads dimensions and distances; this guide uses length, width and height only.

09 Build the central tower

Model build step 9 / 34 | Box



Tinkercad screenshot - observe the tool and the model shape.

W × D × H: 20 × 20 × 108 mm

Find this

A new Box on the base top surface.

Do this

Place a Box on the base, set W 20, D 20, H 108 mm, then press D so it sits on the base.

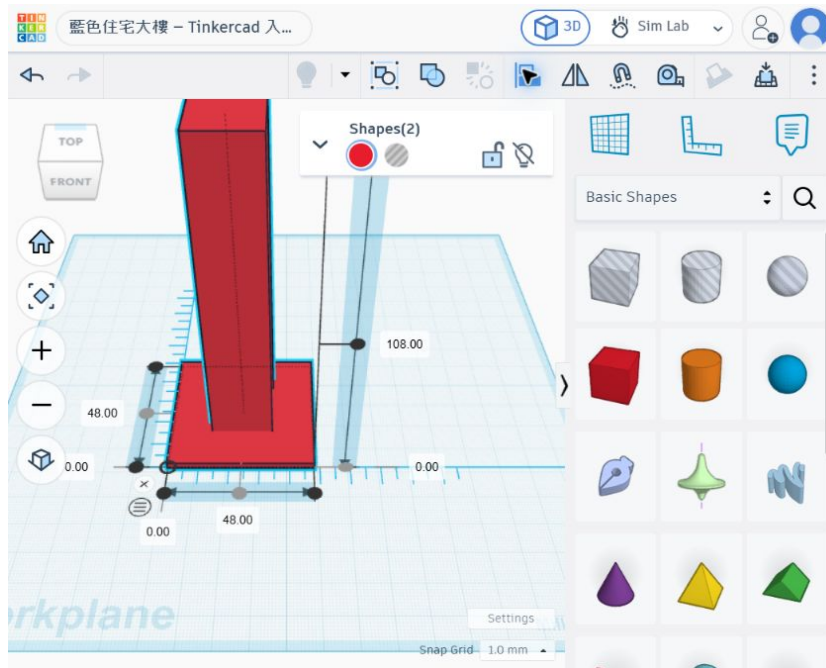
Check

A 20 × 20 × 108 mm tower stands on the base.

Help: Press D after the dimensions are correct so the tower rests on the base.

10 Align the tower to the base centre

Model build step 10 / 34 | Align



Tinkercad screenshot - observe the tool and the model shape.

Find this

The Align tool with the base and tower selected.

Do this

Select the base and tower with Shift. Click Align, then click the two horizontal centre dots. Keep the bottom on the base.

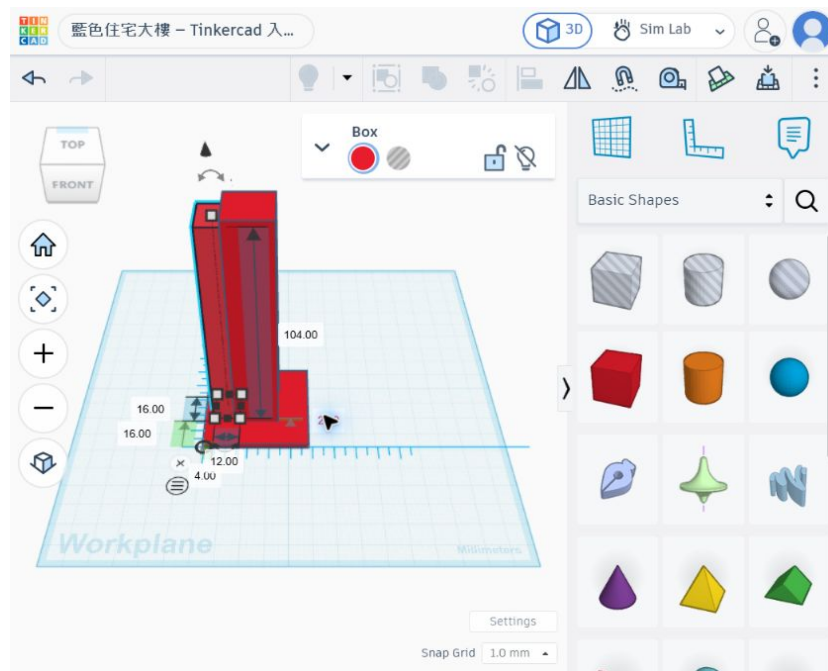
Check

From TOP the tower is centred, and its bottom is still supported.

Help: Align only the two horizontal directions; do not change the tower height.

11 Add the left residential wing

Model build step 11 / 34 | Box



Tinkercad screenshot - observe the tool and the model shape.

W × D × H: 12 × 16 × 104 mm

Find this

A Box beside the central tower.

Do this

Add a Box W 12, D 16, H 104 mm. Press D and move it to the left so it slightly overlaps the tower.

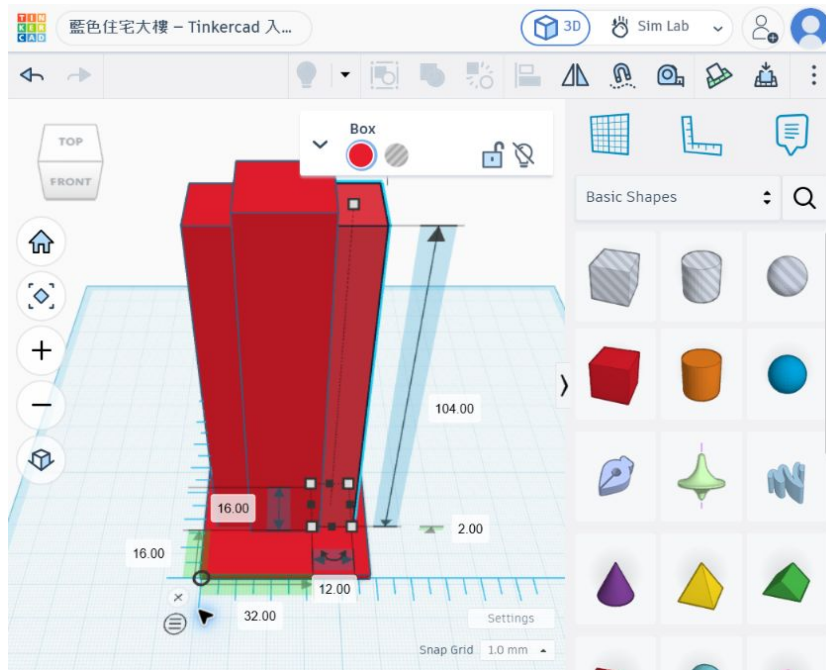
Check

The 12 × 16 × 104 mm left wing touches the tower.

Help: A small overlap prevents a gap in the printed model.

12 Duplicate the wing to the right

Model build step 12 / 34 | Duplicate



Tinkercad screenshot - observe the tool and the model shape.

Find this

The selected wing copy.

Do this

Select the left wing, press Ctrl+D, and move the copy to the right side. Keep the two wings at the same height.

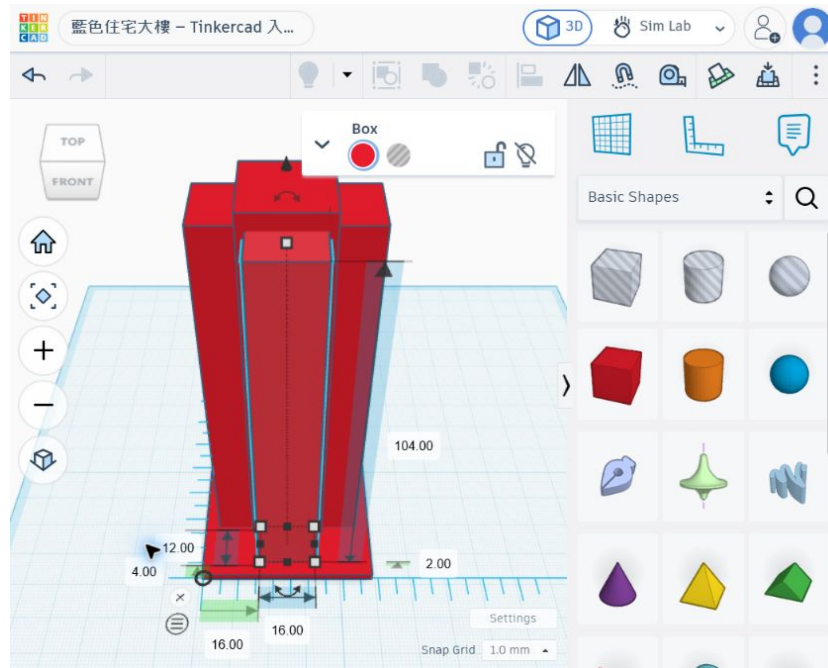
Check

Two matching side wings are visible.

Help: Ctrl+D keeps the copy selected so the next move can be repeated.

13 Rotate a copy for the front wing

Model build step 13 / 34 | Rotate



Tinkercad screenshot - observe the tool and the model shape.

Rotate around vertical axis: 90°; W × D after rotation: 16 × 12 mm

Find this

The curved rotation handle below the selected copy.

Do this

Select the wing copy, rotate it 90 degrees around the vertical axis, then move it to the front. Its W × D becomes 16 × 12 mm.

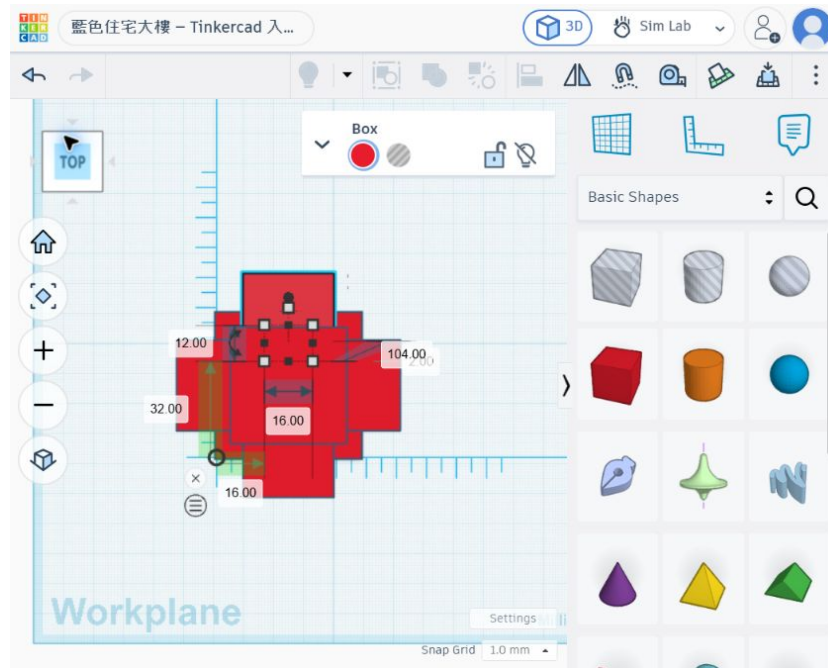
Check

The front wing is turned 90 degrees and extends towards the front.

Help: The curved handle rotates an object; a white square changes its size.

14 Duplicate a rear wing

Model build step 14 / 34 | Duplicate



Tinkercad screenshot - observe the tool and the model shape.

Find this

The selected front wing copy.

Do this

Copy the front wing to the rear, leaving a small overlap. From TOP, the four wings form a cross and remain inside the 48 mm base.

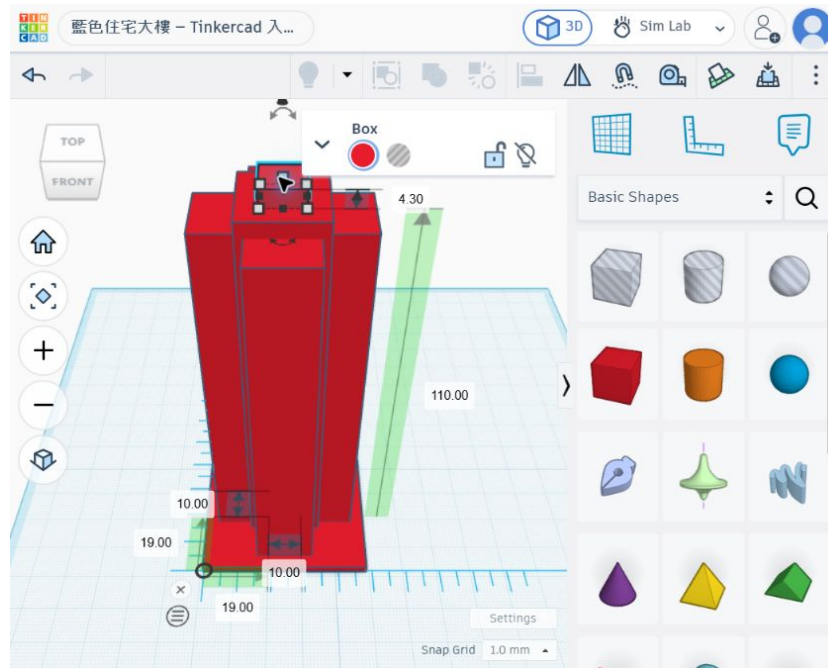
Check

Four wings form a cross within the base.

Help: If the copy falls over, press Ctrl+Z and rotate around the correct axis.

15 Add the roof plant room

Model build step 15 / 34 | Box



Tinkercad screenshot - observe the tool and the model shape.

W × D × H: 10 × 10 × 4.3 mm

Find this

The top surface of the central tower.

Do this

Set Workplane on the tower top. Add a 10 × 10 × 4.3 mm Box, press D and centre it. Total height with the base is $2 + 108 + 4.3 = 114.3$ mm.

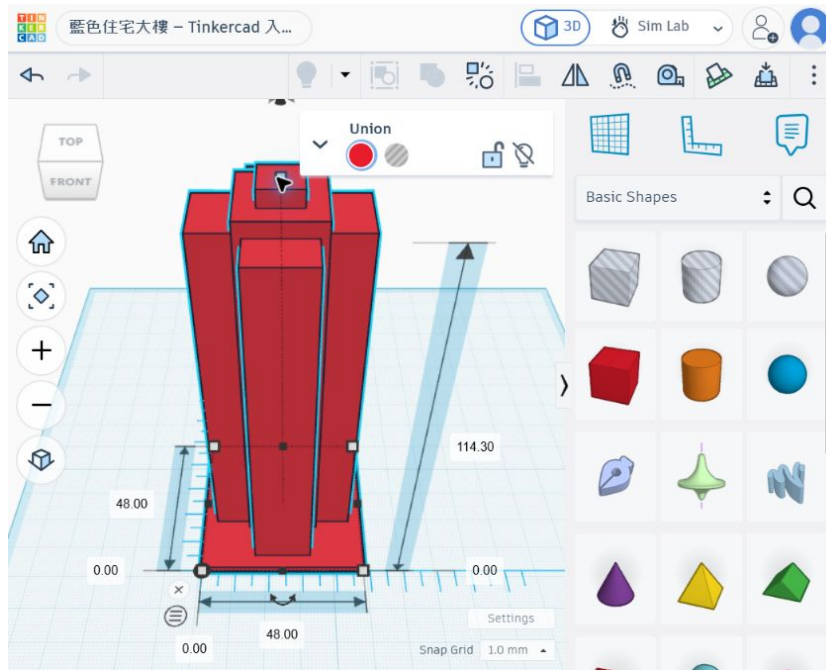
Check

The plant room touches the tower top and the total height is 114.3 mm.

Help: Use Workplane on the top surface, then press D to make the plant room touch it.

16 Group the basic building

Model build step 16 / 34 | Group / Union



Tinkercad screenshot - observe the tool and the model shape.

Find this

All solid building parts.

Do this

Select the connected solid parts and press Ctrl+G (Group / Union).
Check TOP and FRONT.

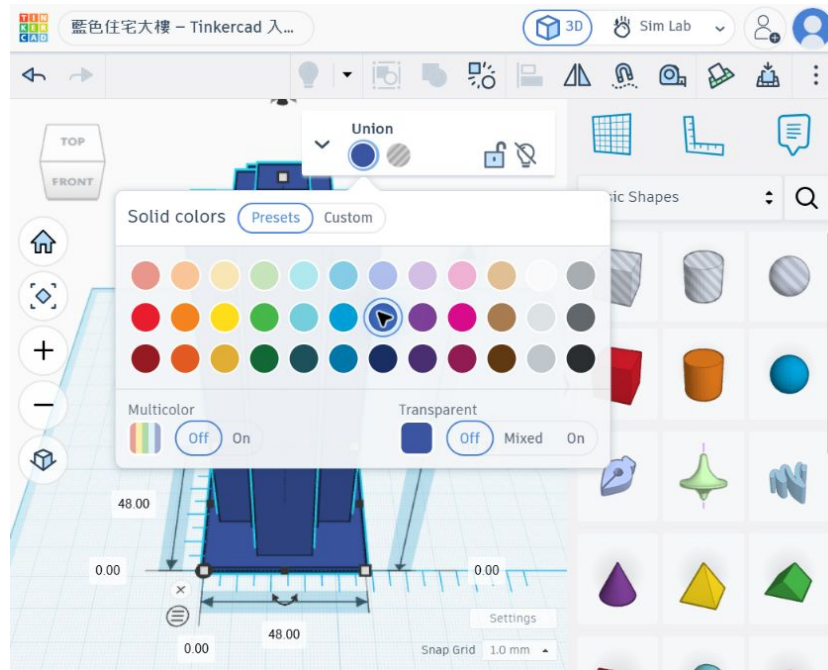
Check

The basic building can be selected as one group.

Help: Keep a copy of the ungrouped design if the teacher wants a version that can be edited.

17 Make the building blue

Model build step 17 / 34 | Solid colour



Tinkercad screenshot - observe the tool and the model shape.

Find this

Shape > Solid colour.

Do this

Select the grouped building, choose Solid colour, and select a deep blue. Click empty space to inspect it.

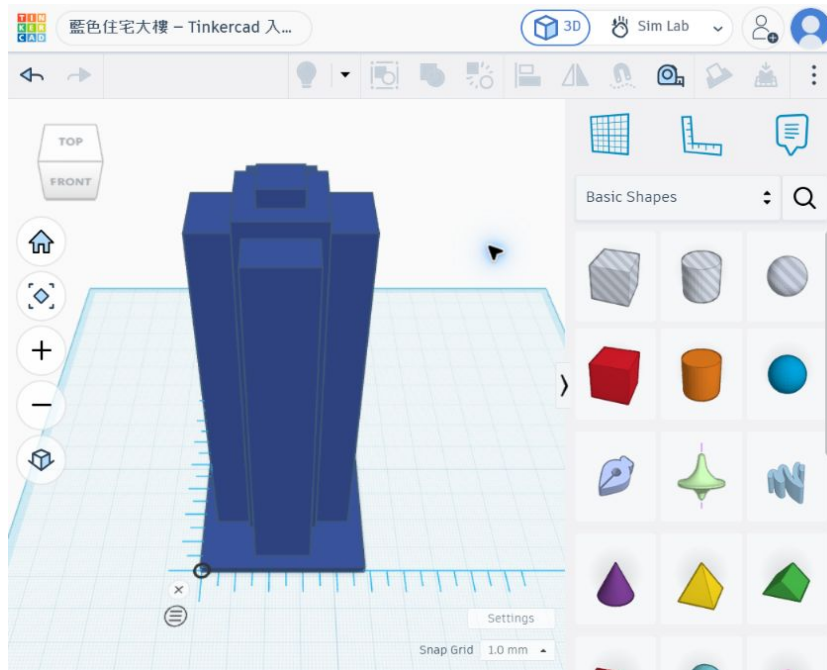
Check

The basic building is blue.

Help: STL does not save colour; blue filament is needed for a blue print.

18 Finish the basic version before windows

Model build step 18 / 34 | Home view



Tinkercad screenshot - observe the tool and the model shape.

Base: $48 \times 48 \times 2$ mm; Total height: 114.3 mm

Find this

The completed basic building before windows.

Do this

Use Home to view the finished basic building. Save it before starting the windows.

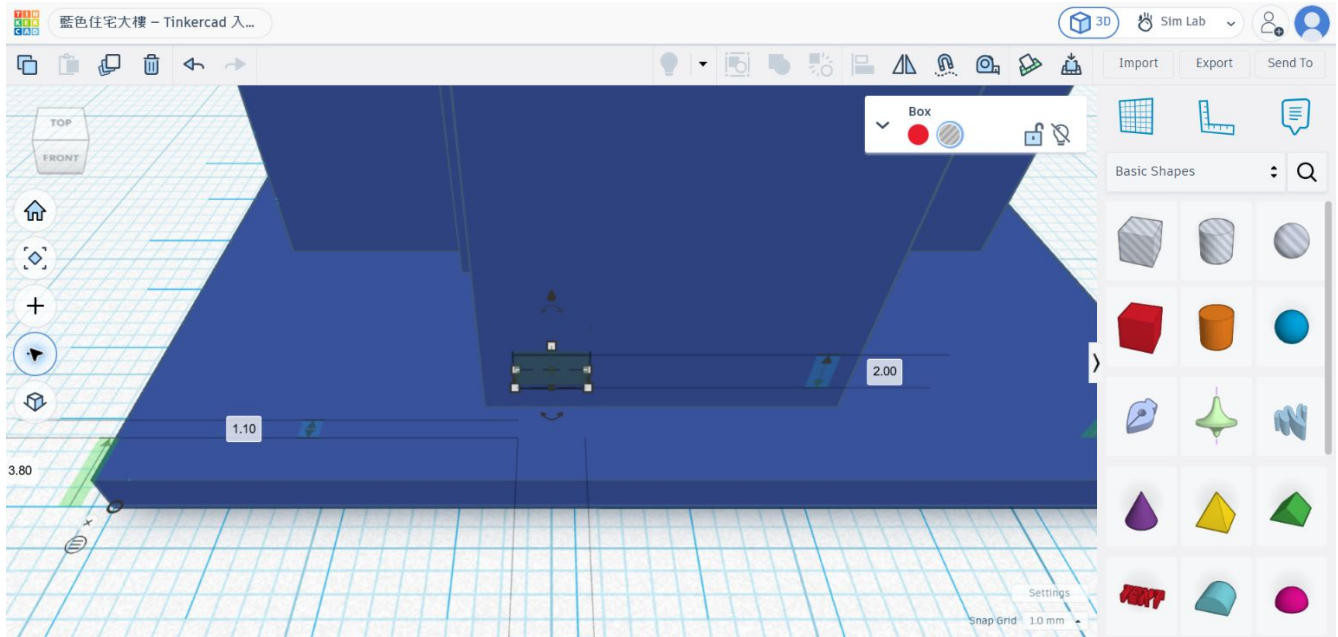
Check

The shape is ready for window work.

Help: If time is short, export the basic version first and add windows later.

19 Make the first window recess

Model build step 19 / 34 | Hole



Tinkercad screenshot - observe the tool and the model shape.

W × D × H: 3.2 × 1.1 × 2 mm; Wall overlap: 0.9 mm

Find this

A small Box changed to Hole.

Do this

Drag in a Box, set W 3.2, D 1.1, H 2 mm, change it to Hole and overlap the front wing wall.

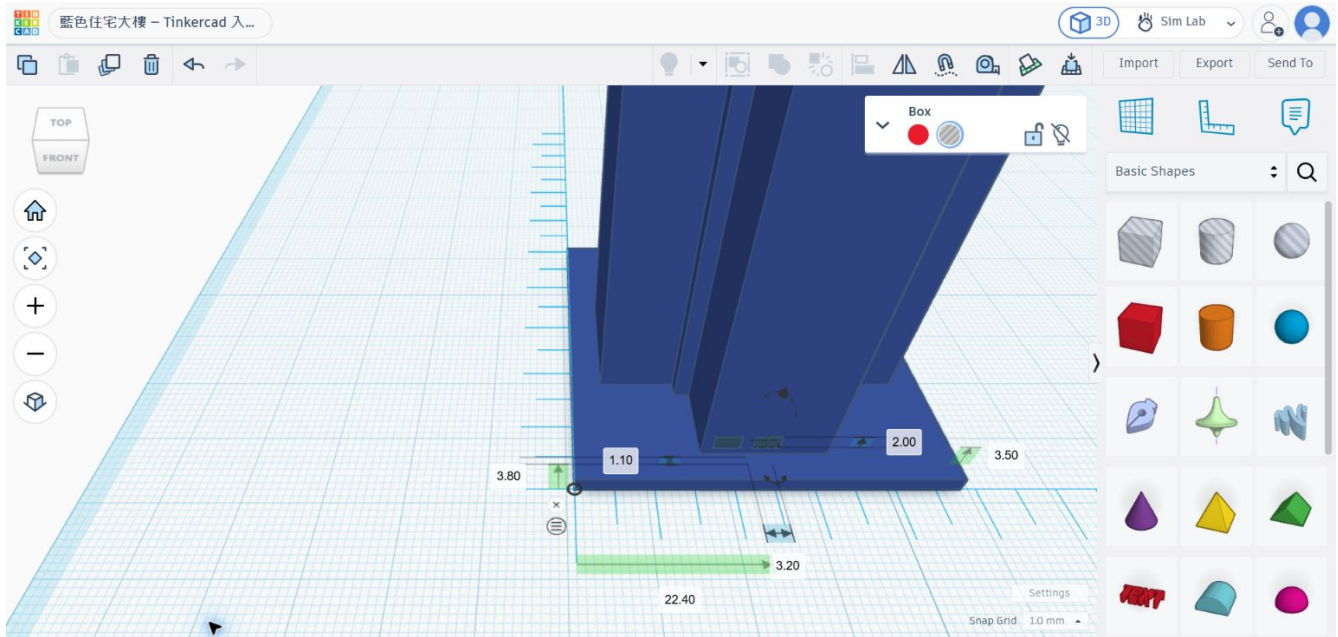
Check

A shallow Hole overlaps the wall.

Help: A Hole must overlap the wall. A Hole floating beside it will not cut anything.

20 Duplicate the second window

Model build step 20 / 34 | Duplicate



Tinkercad screenshot - observe the tool and the model shape.

Horizontal spacing: 5 mm

Find this

The selected first window hole.

Do this

Set Snap Grid to 1.0 mm. Copy the first Hole and move it right by 5 mm. Keep the copy selected; the clear space between windows is 1.8 mm.

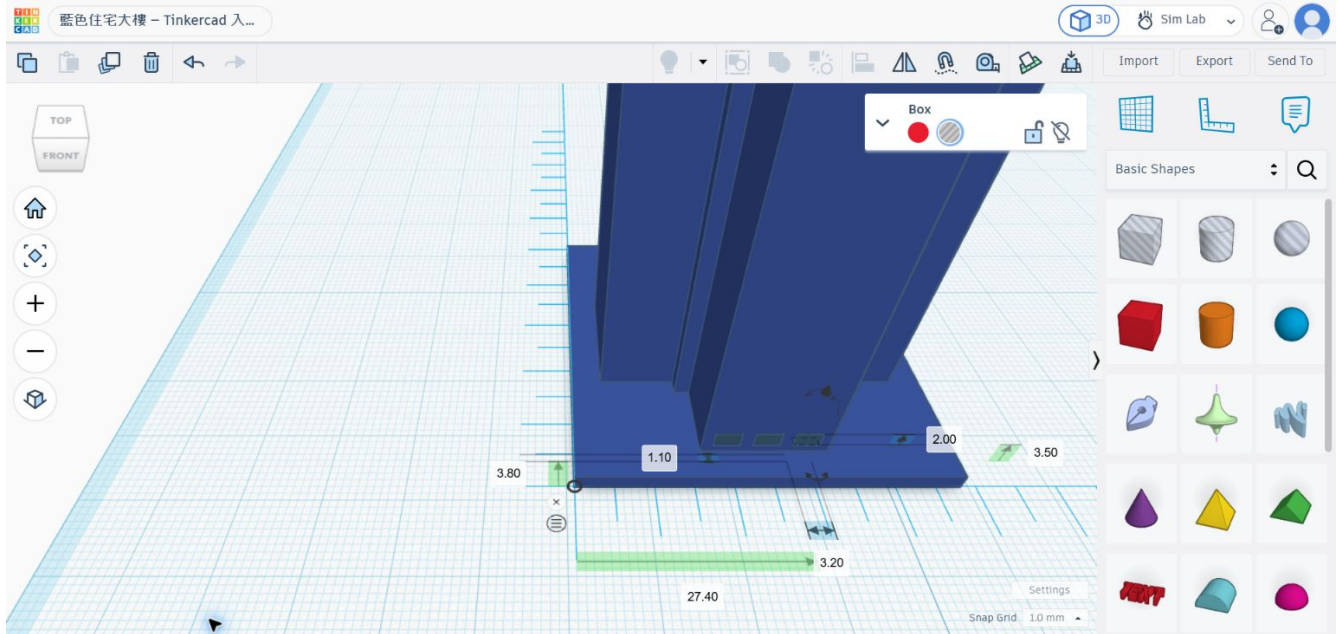
Check

Two equal-height windows are 5 mm apart from matching edges.

Help: Set Snap Grid to 1.0 mm before making repeated windows.

21 Repeat the transform to make three windows

Model build step 21 / 34 | Duplicate



Tinkercad screenshot - observe the tool and the model shape.

Find this

The selected copied window.

Do this

Press Ctrl+D again to repeat the 5 mm horizontal move. Make a row of three holes.

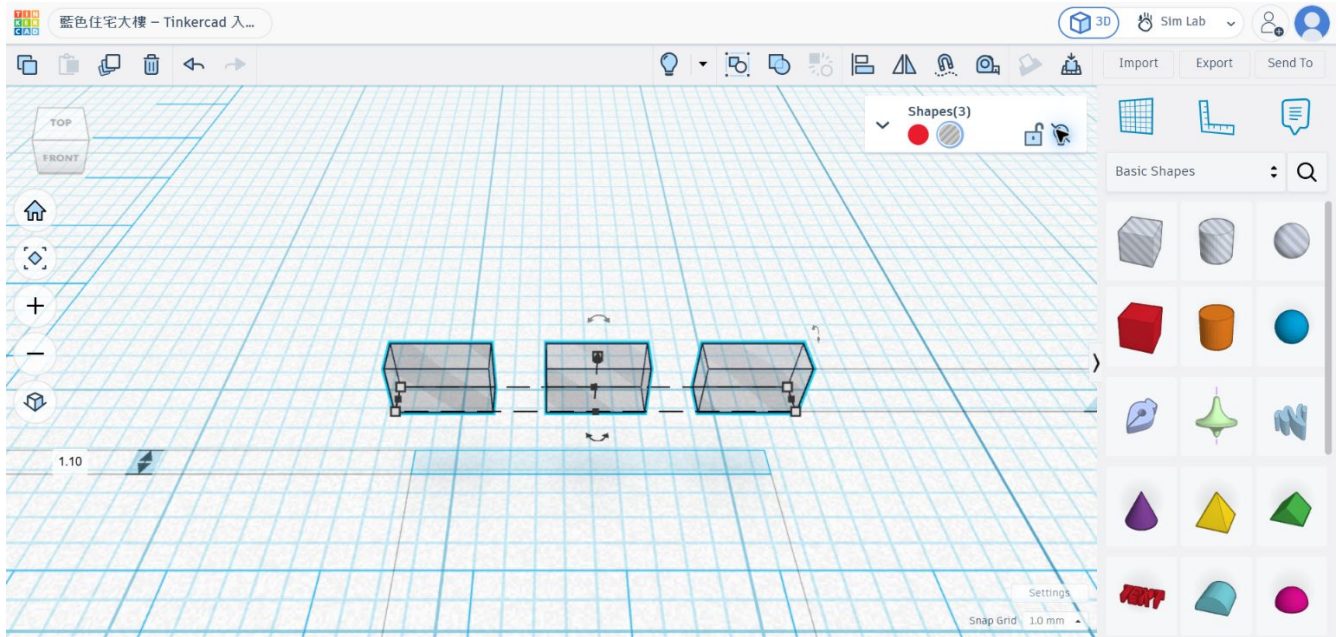
Check

Three holes form one straight row.

Help: 5 mm is the copy movement; the clear wall space between holes is smaller.

22 Select one row of holes

Model build step 22 / 34 | Workplane



Tinkercad screenshot - observe the tool and the model shape.

Find this

The first row of three holes.

Do this

Select only the first row of holes. If needed, hide the building with Ctrl+H while you work.

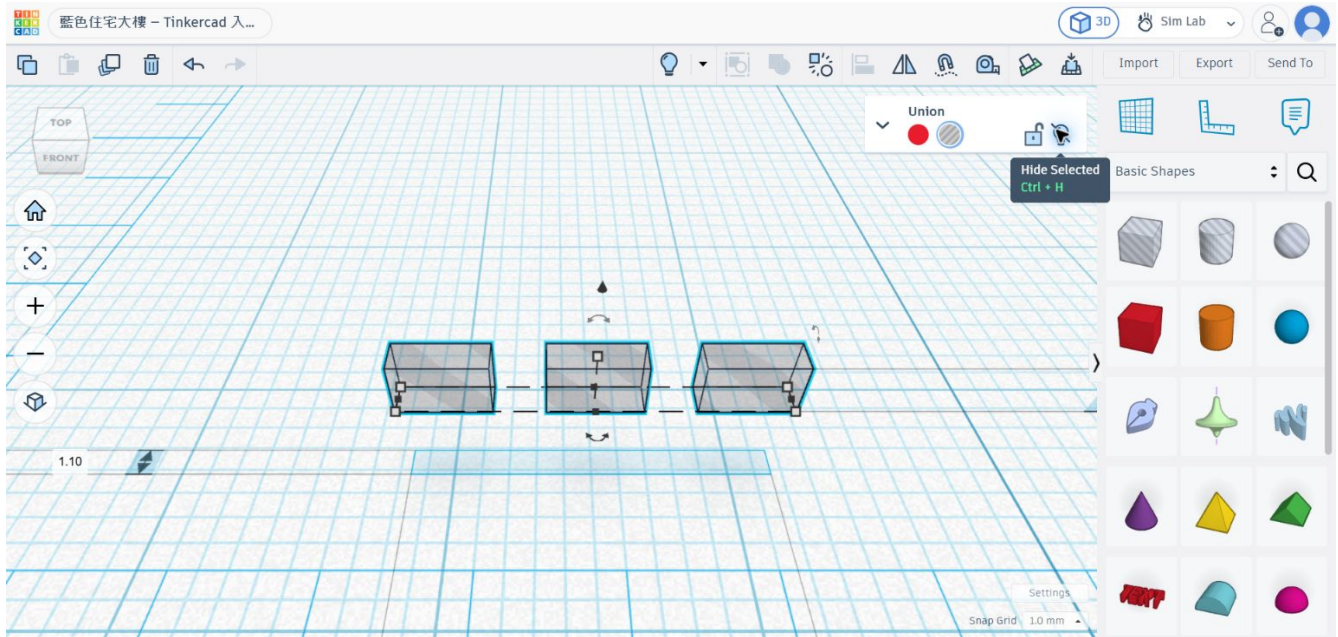
Check

Only the intended row is selected.

Help: Select only the intended row before pressing Ctrl+G.

23 Group one row of holes

Model build step 23 / 34 | Group / Union



Tinkercad screenshot - observe the tool and the model shape.

Row bounding box $W \times D \times H$: $13.2 \times 1.1 \times 2$ mm

Find this

The row of holes selected together.

Do this

Press Ctrl+G to group the selected row of holes. Do not group it with the building yet.

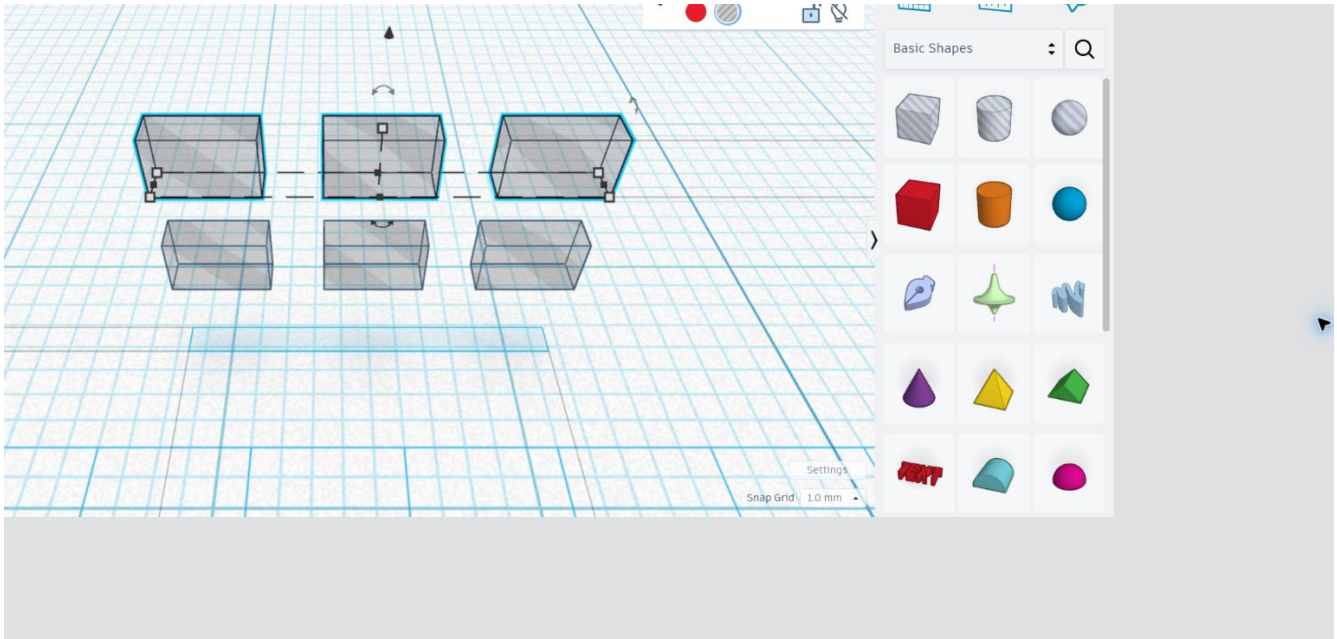
Check

The row is one grouped Hole object.

Help: Grouped holes can still be copied and moved as one row.

24 Duplicate the second window row

Model build step 24 / 34 | Duplicate



Tinkercad screenshot - observe the tool and the model shape.

Rise per row: 4 mm; Hole height H: 2 mm

Find this

The grouped first row of holes.

Do this

Press Ctrl+D to copy the row, then use Ctrl plus the Up arrow four times to raise it 4 mm.

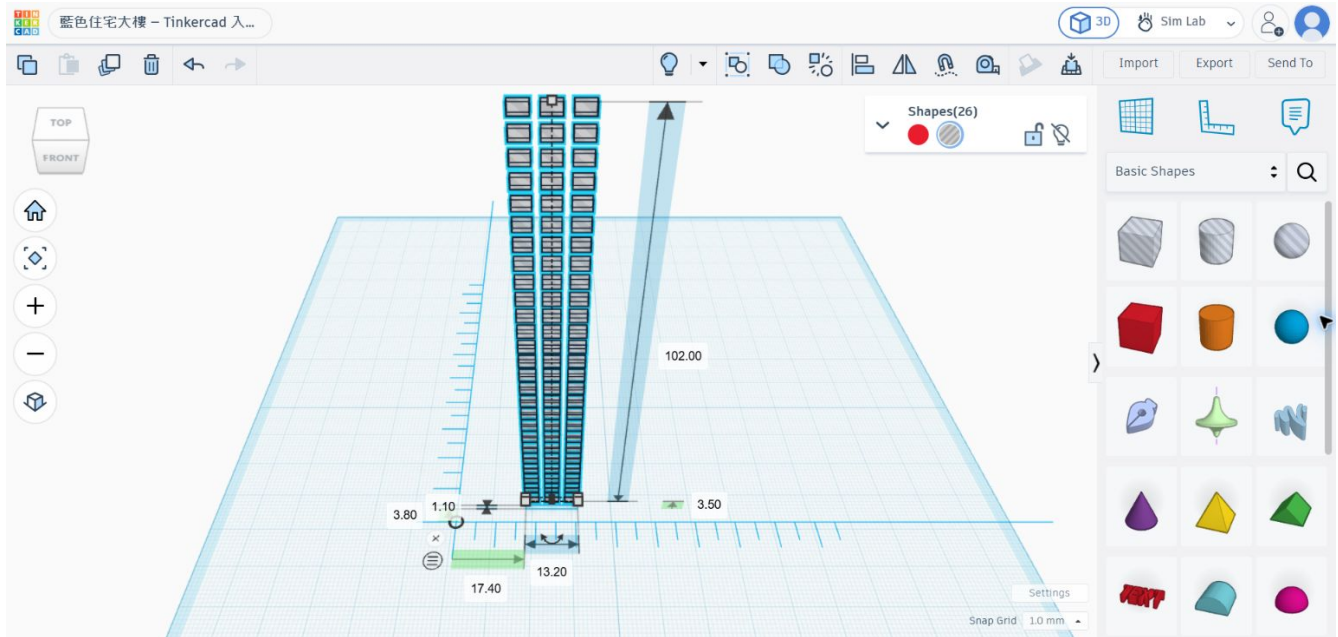
Check

Two equal rows are visible.

Help: Use Ctrl plus the arrow key to move a copy; do not stretch the Hole itself.

25 Repeat to make twenty-six rows

Model build step 25 / 34 | Duplicate



Tinkercad screenshot - observe the tool and the model shape.

Total rows: 26 rows; Further duplicates: 24 times

Find this

The second window row.

Do this

Repeat the row until there are 26 rows. There are 3 windows per row, so the face contains 78 holes.

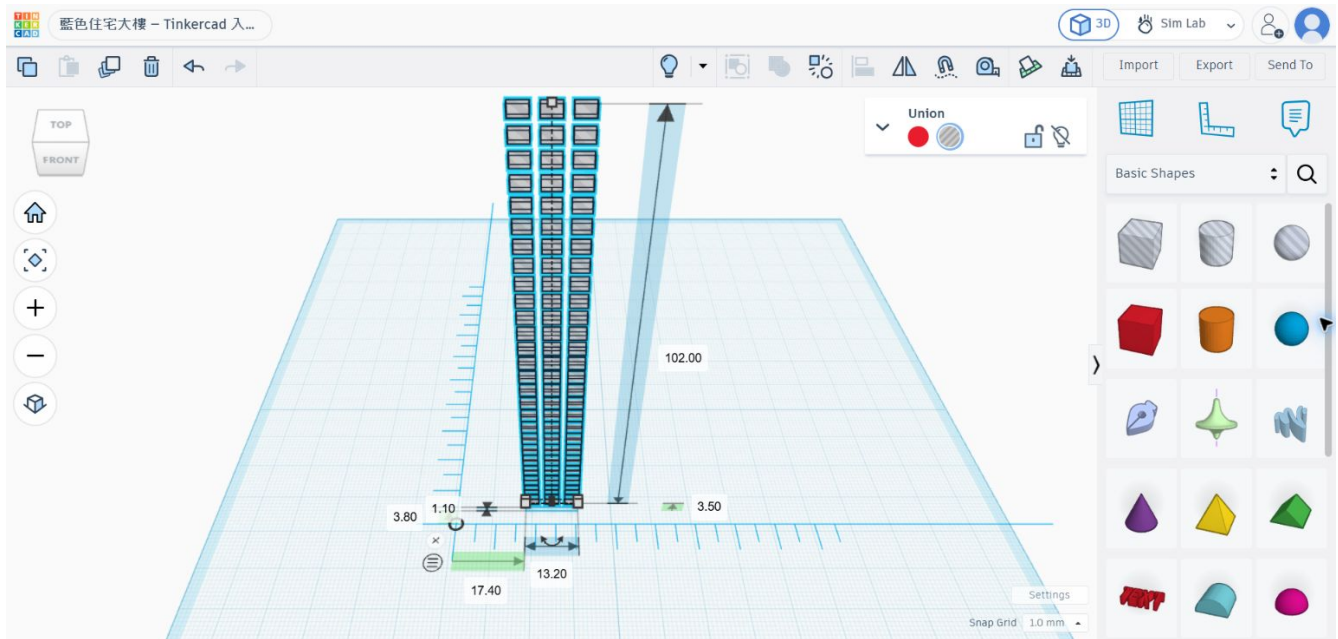
Check

The 26 rows remain inside the wing wall.

Help: Keep the highest row below the roof edge.

26 Group one complete face

Model build step 26 / 34 | Group / Union



Tinkercad screenshot - observe the tool and the model shape.

Find this

The repeated rows on one face.

Do this

Select the complete face and group it as one Hole object. The grouped face represents 78 window holes.

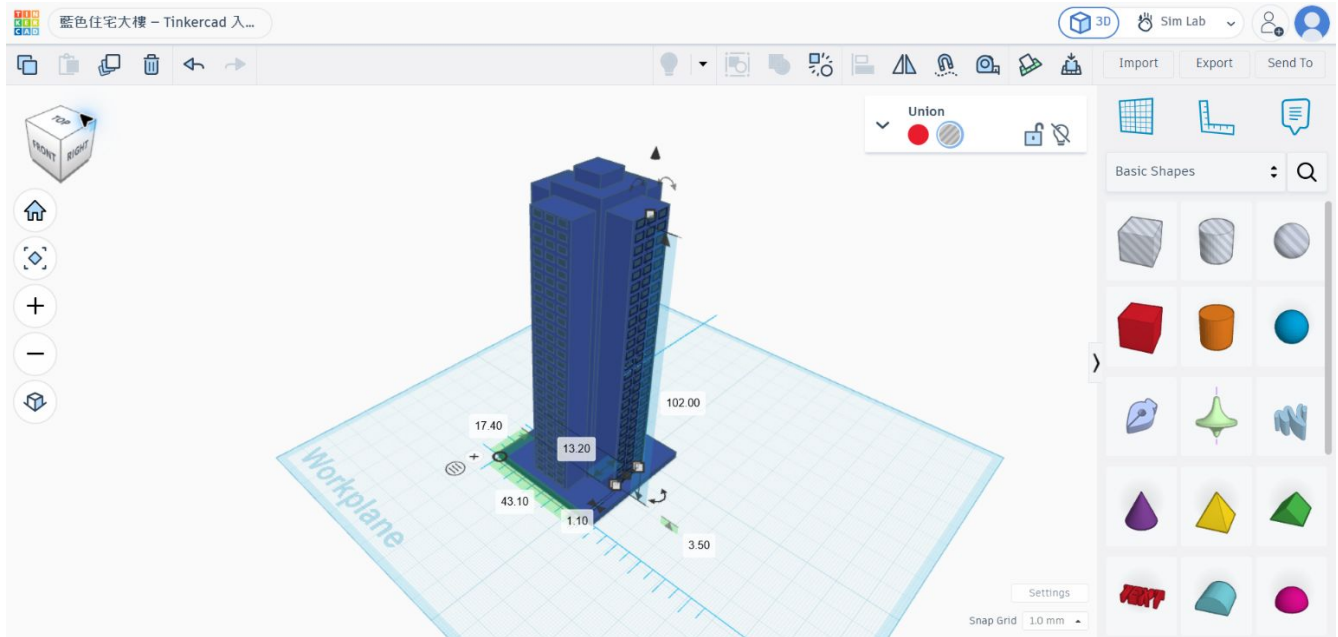
Check

One grouped face of 78 holes is selected.

Help: Do not group the holes with the building until all four faces are ready.

27 Copy the holes to the right facade

Model build step 27 / 34 | Rotate



Tinkercad screenshot - observe the tool and the model shape.

Find this

The complete grouped face of holes.

Do this

Copy the face group, rotate it 90 degrees, and move it to the right wing. Show all to check overlap.

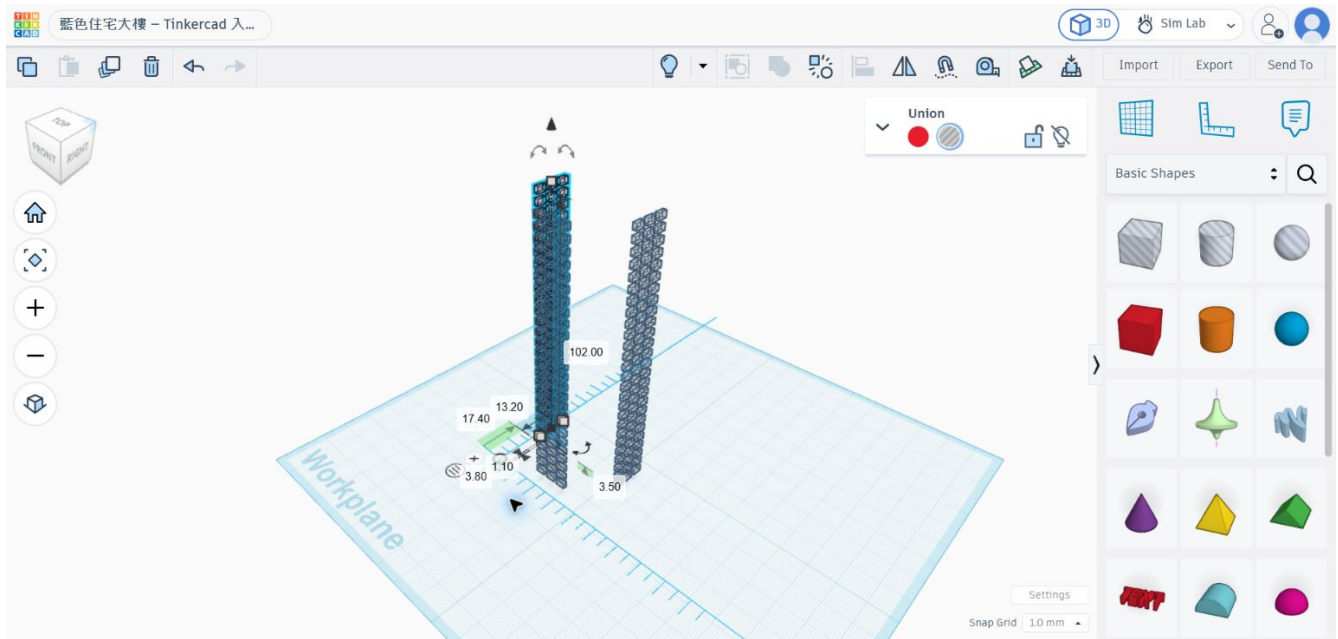
Check

The right wing has matching recessed windows.

Help: After rotating a face, move it to the new wall and check it in TOP view.

28 Copy the holes to the left facade

Model build step 28 / 34 | Duplicate



Tinkercad screenshot - observe the tool and the model shape.

Find this

The right facade after rotation.

Do this

Copy the face group to the left wing. Keep its height aligned with the right side.

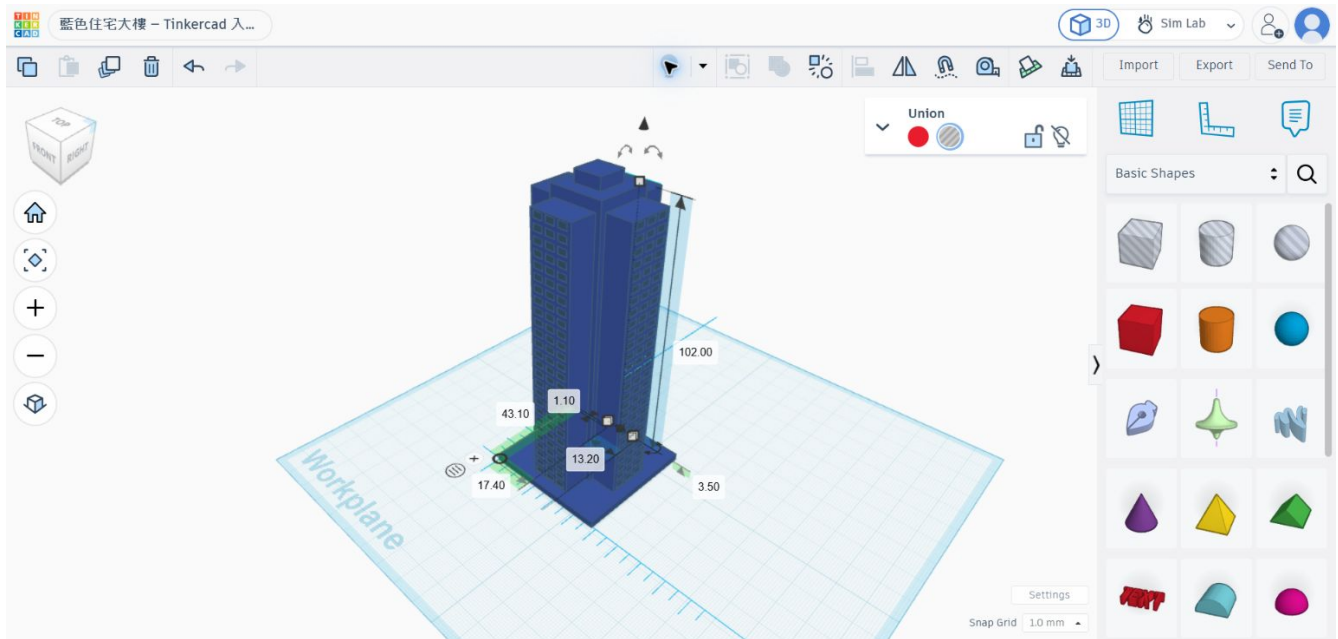
Check

The left wing has matching recessed windows.

Help: Keep the same height as the opposite wing.

29 Copy the holes to the rear facade

Model build step 29 / 34 | Duplicate



Tinkercad screenshot - observe the tool and the model shape.

Find this

The left facade copy.

Do this

Copy the original front face to the rear. With four faces, there are $4 \times 78 = 312$ window holes; check W 13.2 and D 1.1 mm.

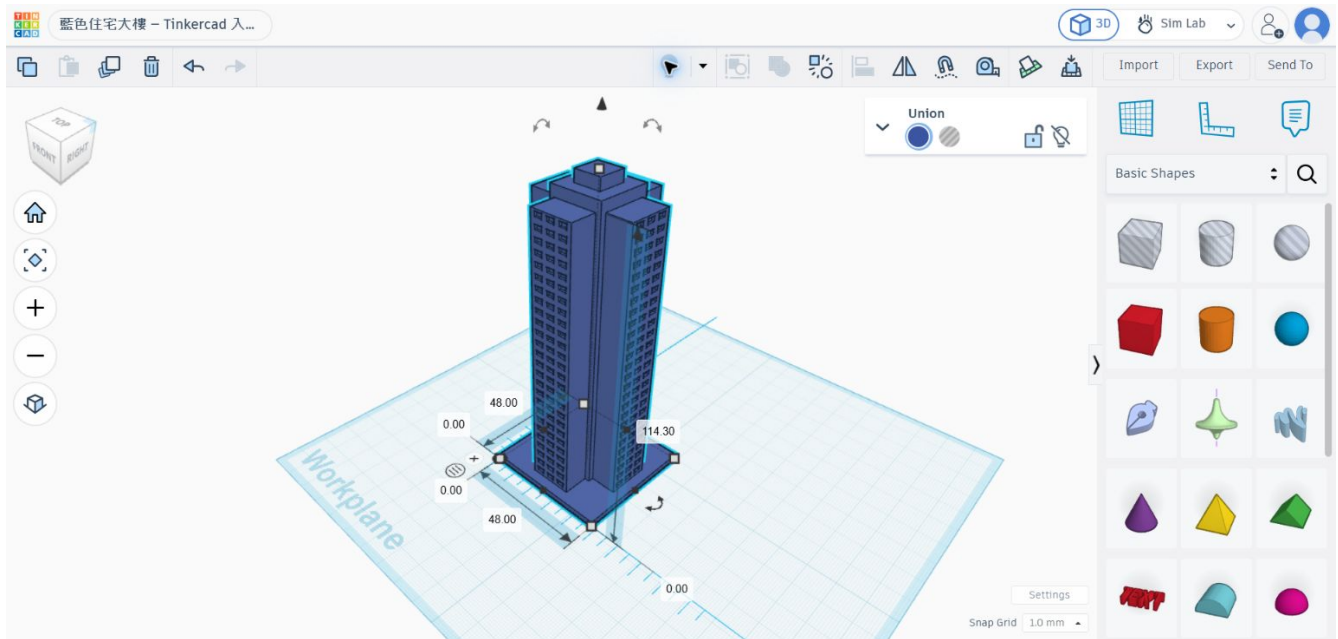
Check

The rear copy overlaps the rear wall and all four faces match.

Help: Use TOP and the rear view to check the wall overlap.

30 Cut all four window faces at once

Model build step 30 / 34 | Group / Union



Tinkercad screenshot - observe the tool and the model shape.

Find this

The rear facade copy.

Do this

Select the building and the four hole groups, then press Ctrl+G (Union). If Shapes(5) appears, wait for the cut to finish.

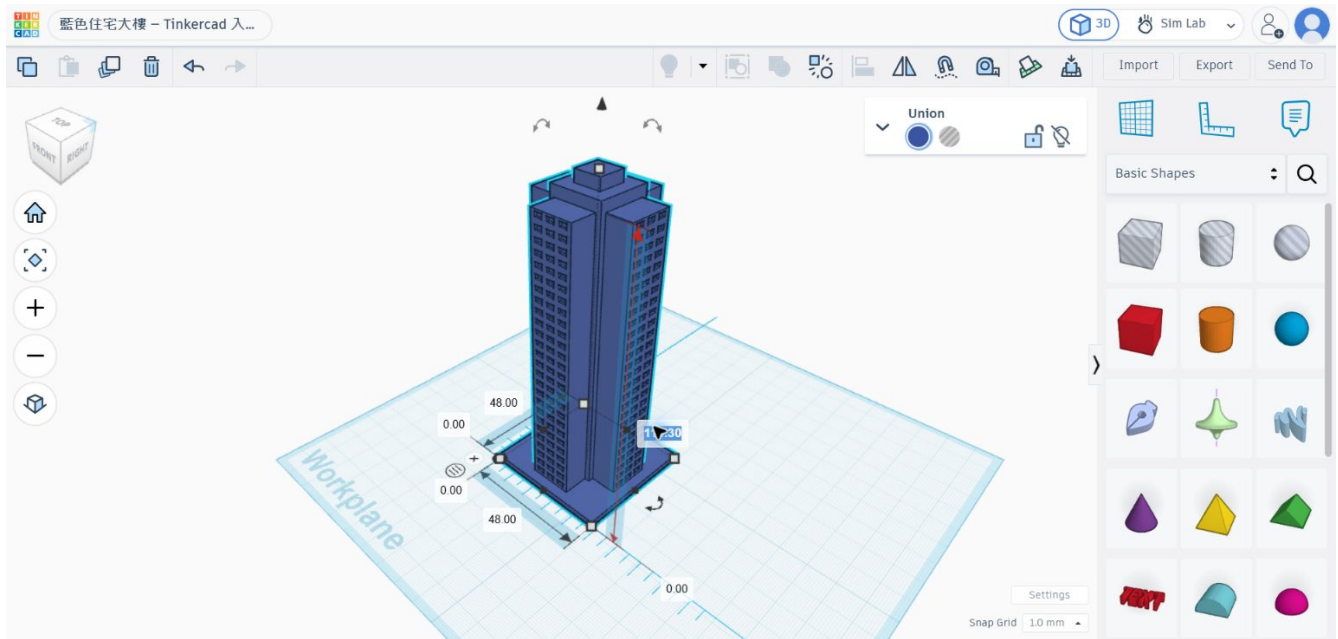
Check

The four facades show rows of shallow recessed windows.

Help: Select the building and holes together only at the final Union step.

31 Check size and connections

Model build step 31 / 34 | Ruler



Tinkercad screenshot - observe the tool and the model shape.

W × D × H: 48 × 48 × 114.3 mm

Find this

The building and all window-hole groups.

Do this

Use Ruler and the four views to confirm the base, tower, wings and plant room are connected.

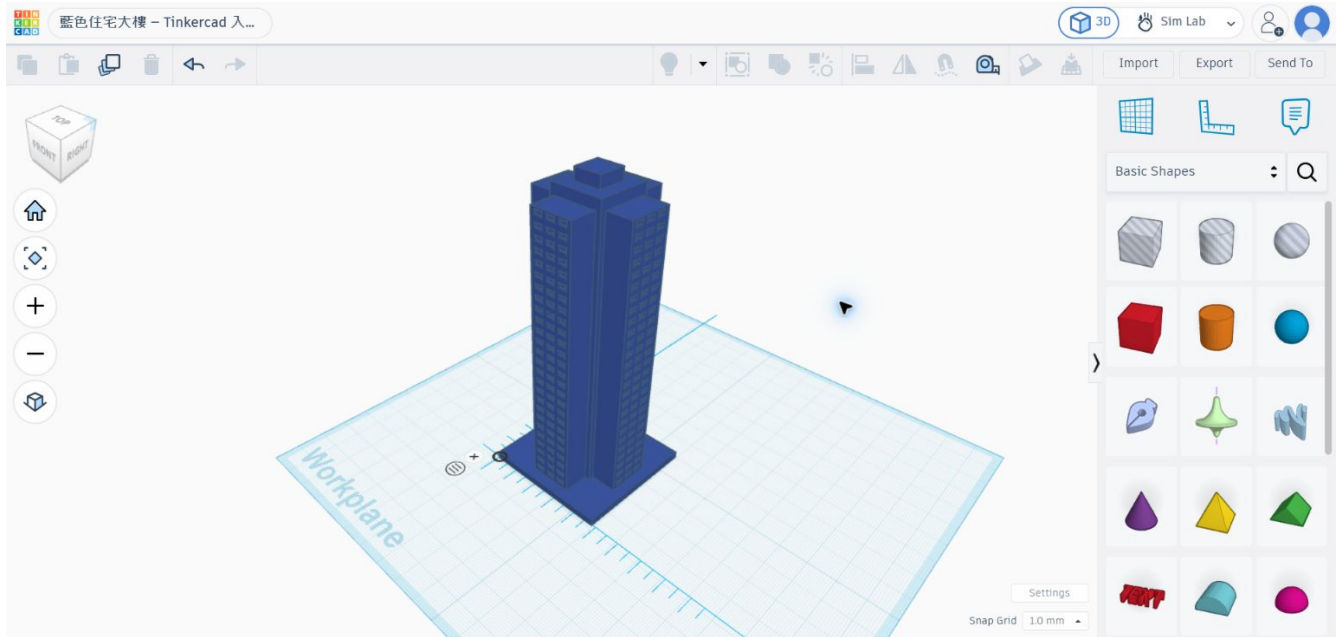
Check

W 48, D 48, H 114.3 mm and no separated parts.

Help: If a part looks detached, undo and check the Workplane and D key.

32 Confirm the blue finish

Model build step 32 / 34 | Solid colour



Tinkercad screenshot - observe the tool and the model shape.

Find this

Ruler and the four connected facades.

Do this

Select the finished solid and set Solid colour to deep blue. Click empty space and press Home.

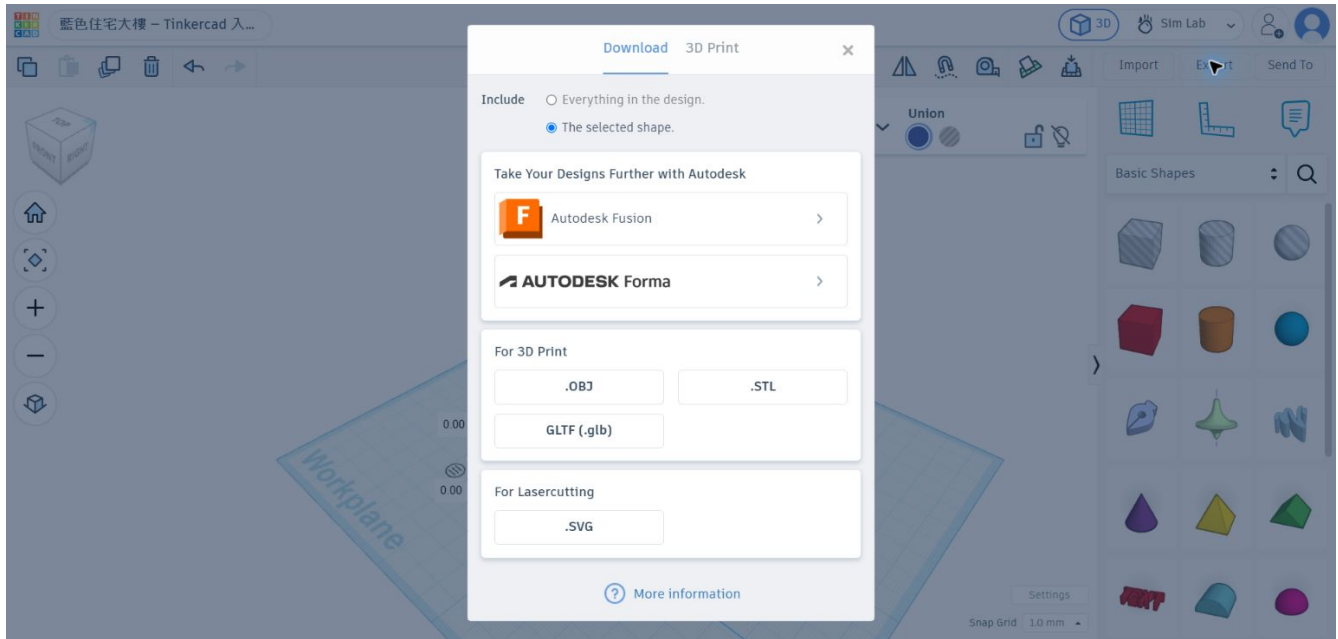
Check

The completed model is blue with clear recesses.

Help: The on-screen blue may differ from printed filament colour.

33 Export an STL from Tinkercad

Model build step 33 / 34 | Export



Tinkercad screenshot - observe the tool and the model shape.

Find this

The selected building and Solid colour control.

Do this

Select the complete building, click Export, choose The selected shape if asked, then choose .STL.

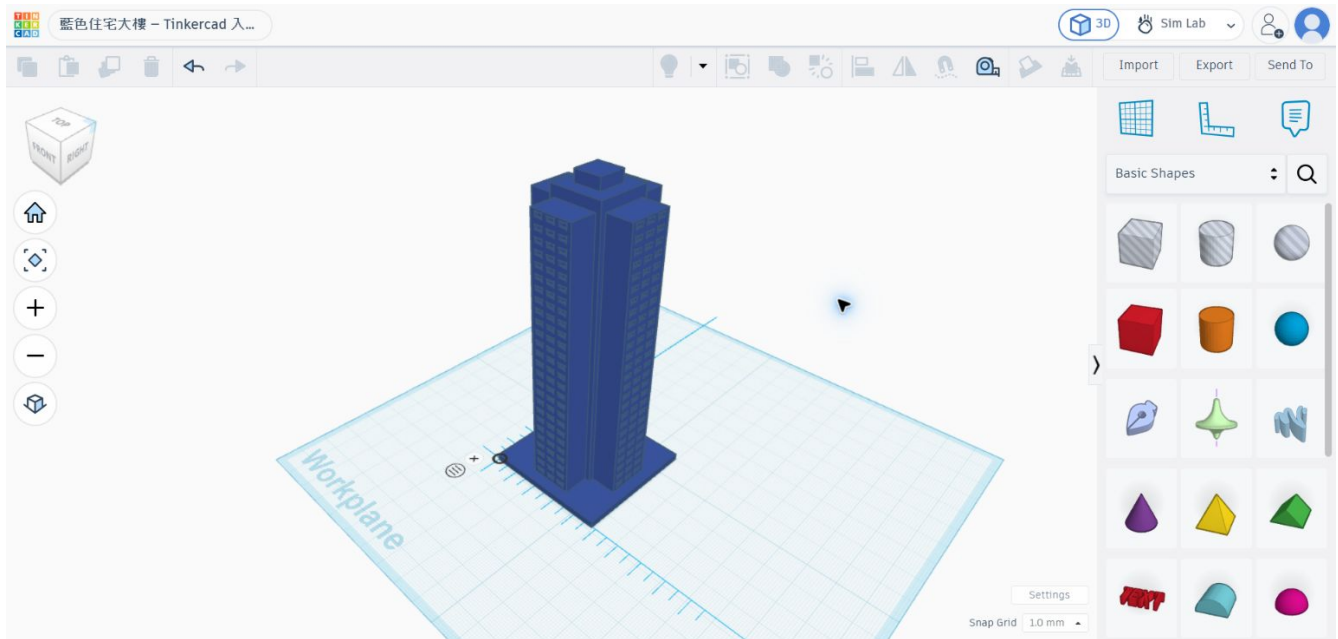
Check

An .stl file appears in Downloads.

Help: Choose The selected shape when Tinkercad asks which object to export.

34 Give the file to the teacher for printing

Model build step 34 / 34 | Export



Tinkercad screenshot - observe the tool and the model shape.

Delivery file: .STL; Import units: mm; Model total height: 114.3 mm

Find this

Export at the top right of Tinkercad.

Do this

Give the STL to the teacher. Confirm millimetres, base down, total height 114.3 mm, and inspect the slice preview.

Check

The teacher can check and prepare the model for printing.

Help: The teacher checks wall thickness and window clarity in the slicer preview.